

Georg von Westphalen

MEDKAYFAB: PHARMACOLOGY CORE EDITION

UNCLE FRIZZLEBOB'S KRAZY KAYFABIZZARO KOMIK KARD KULT KLUB!



Roxy Receptor

Receptor Theory

STAT
K_d
(Dissociation
Constant)



POWER: Occupancy Multiplier
LORE: Partial agonists act as functional antagonists in high-efficacy systems.

Georgie GPCR

G-Protein Coupled Receptors

STAT
34% (Target of all
approved drugs)



POWER: G-Protein Split
LORE: $G_s = \uparrow cAMP$, $G_i = \downarrow cAMP$, $G_q = IP3/DAG$.

Kinny Kinase Receptor

RTKs

STAT:
1 Transmembrane
Domain



POWER: Dimerize & Phosphorylate
LORE: Imatinib selectively targets BCR-ABL without killing all cellular tyrosine kinase activity.

Nicky Nuclear

Nuclear Receptors

STAT:
Hours to Days
(Onset)



POWER: Gene Rewrite
LORE: Corticosteroid withdrawal must be gradual due to profound HPA axis suppression.

BLÖDSINN!

Iggy Ion Channel

Ligand-Gated

STAT
Milliseconds



POWER: Ion Flux
LORE: Succinylcholine causes initial fasciculations before flaccid paralysis.

Volta the Voltage-Gated

Voltage-Gated

STAT
pH < 7.4
(Inflamed tissue)



POWER: Use-Dependent Block
LORE: Local anesthetics fail in acidic tissue because they ionize and cannot cross the membrane.

Beta the Beta-Arrestin

Desensitization

STAT:
 $\beta 2$ (Target for downregulation)



POWER: Endocytosis
LORE: Salbutamol overuse causes rapid receptor downregulation (tachyphylaxis).

Second the Second Messenger

cAMP/IP3

STAT:
PDE5 (Sildenafil target)



POWER: Relay Cascade
LORE: Sildenafil inhibits PDE5 $\rightarrow$ cGMP accumulation $\rightarrow$ profound vasodilation.

BLÖDSINN!



KayfaBINGO!



9 Allo the Allosteric

Allosteric Modulation

STAT
EC₂₀
Shift



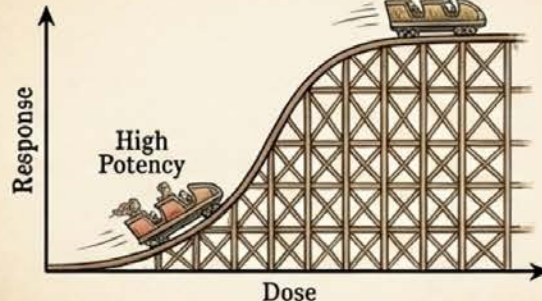
POWER: Affinity Modifier
LORE: Benzodiazepines require GABA presence to work—a ceiling effect making them safer than barbiturates.

10 Dose the Dose-Response

EC₅₀/E_{max}

High Efficacy

STAT
Potency
vs.
Efficacy



POWER: The Therapeutic Window
LORE: Potency (EC₅₀) does not equal Efficacy (E_{max}).



11 Abby Absorption

Bioavailability

STAT
 $F = \frac{AUC(oral)}{AUC(iv)}$



POWER: First-Pass Evasion
LORE: Glyceryl trinitrate is given sublingually to bypass massive hepatic destruction.

12 First Pass Frank

Hepatic First-Pass

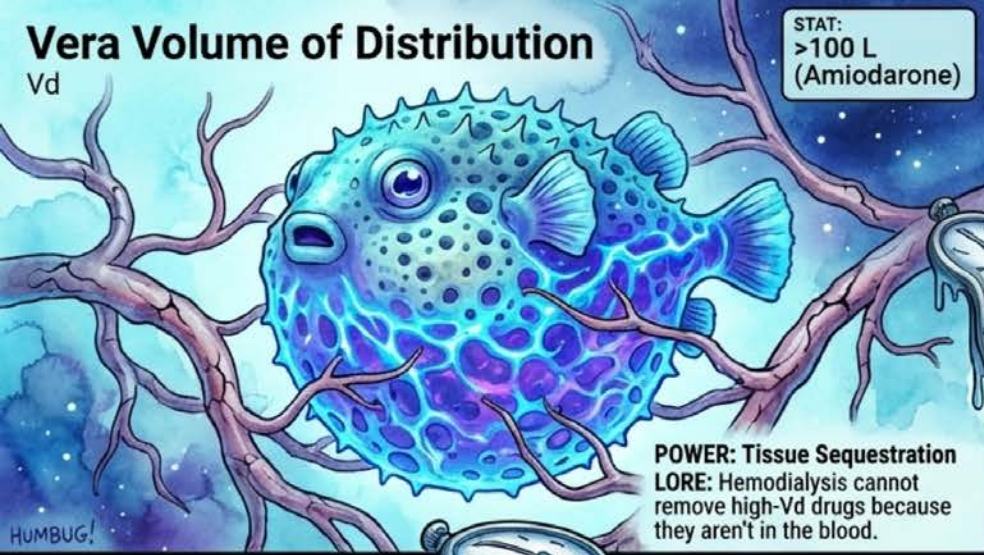
STAT
>70%
Extraction
Ratio



POWER: Portal Shredder
LORE: Codeine is a prodrug; without CYP2D6 metabolism, it provides zero analgesia.



← Stupid Cliffs!



Vera Volume of Distribution

Vd

STAT:
>100 L
(Amiodarone)

POWER: Tissue Sequestration
LORE: Hemodialysis cannot remove high-Vd drugs because they aren't in the blood.

HUMBUG!



Protein Priya

Plasma Protein Binding

STAT:
99% Bound
(Warfarin)

POWER: Plasma Hoarding
LORE: Hypoalbuminemia leads to a dangerous spike in the free fraction of highly bound drugs like phenytoin.



Hal the Half-Life

$t_{1/2}$

STAT:
 $\sim 5 \times t_{1/2}$ (Steady state)

POWER: Exponential Decay
LORE: Half-life depends on BOTH volume of distribution and clearance.

HUMBUG!



Zero the Zero-Order

Kinetics

STAT:
Constant Rate

POWER: Saturation Cap
LORE: In zero-order kinetics, a small dose increase causes a massive, **GW** Georg v. Westphalen

17 Clara Clearance

Hepatic/Renal

STAT

eGFR < 30
(Metformin danger)



POWER: Organ Extraction

LORE: Severe renal impairment + Metformin = catastrophic lactic acidosis.

18 Cyp the CYP450 Chief

Enzyme Overview

STAT

50%
(CYP3A4 share)



POWER: Metabolic Sorting

LORE: Fluconazole inhibits CYP2C9 → massive Warfarin INR spike.

19 Indie the Inducer

Enzyme Induction

STAT

Days to Weeks
(Onset)



POWER: Enzyme Upregulation (PHORCEPS)

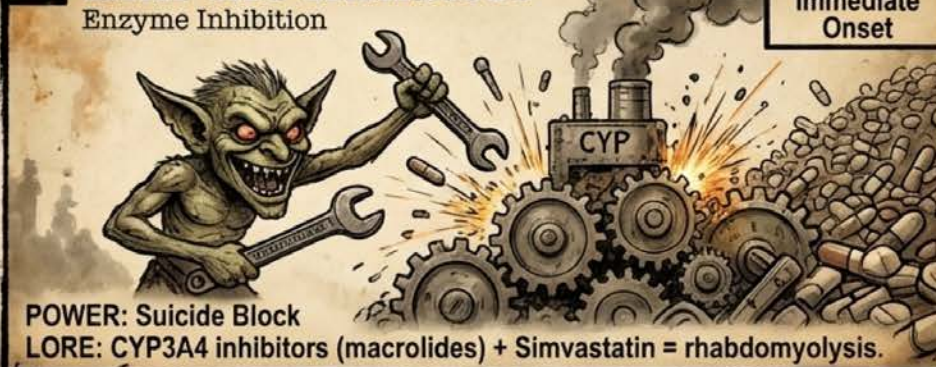
LORE: Rifampicin induction destroys oral contraceptive efficacy, leading to unintended pregnancy.

20 Inhi the Inhibitor

Enzyme Inhibition

STAT

Immediate
Onset



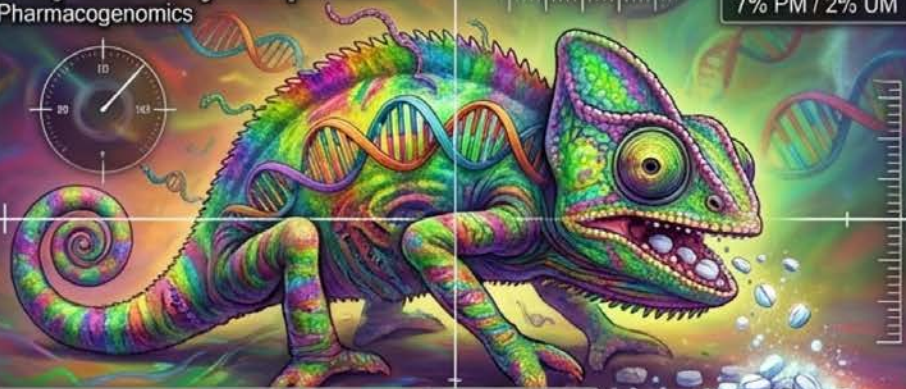
POWER: Suicide Block

LORE: CYP3A4 inhibitors (macrolides) + Simvastatin = rhabdomyolysis.

What the FLUFF?!

Poly the Polymorphism

Pharmacogenomics



STAT

7% PM / 2% UM

POWER: Phenotype Roulette

LORE: Ultrarapid metabolizers turn codeine into fatal doses of morphine, especially dangerous to breastfed neonates.

Sym the Sympathetic

SNS



STAT

80% Epi / 20% NE

POWER: Fight or Flight

LORE: Low-dose epinephrine causes β_2 vasodilation; high-dose shifts to massive α_1 vasoconstriction.

Thera the TDM

Therapeutic Drug Monitor



STAT

>2.0 mEq/L
(Lithium Toxicity)

POWER: Narrow Therapeutic Index

LORE: Draw digoxin levels at least 6 hours post-dose to avoid falsely toxic tissue-distribution readings.

Para the Parasympathetic

PNS



STAT

M1, M2, M3

POWER: Rest & Digest

LORE: Anticholinergic toxicity in the elderly causes confusion, dry mouth, and urinary retention.

25 Nicky NMJ
Neuromuscular Junction

STAT
5-Subunit nAChR



POWER: Vesicle Fusion
LORE: Neostigmine reverses rocuronium paralysis, but cannot reverse depolarizing succinylcholine.

27 Beta Barry
Beta Blockers

STAT:
 β_1 vs β_2



POWER: Cardioselectivity
LORE: Non-selective beta blockers are dangerous in asthma; cardioselective (metoprolol) are preferred.

26 Alpha Annie
Alpha Blockers

STAT
 α_1 vs α_2



POWER: Vascular Blockade
LORE: Alpha-blockade (phenoxybenzamine) **MUST** precede beta-blockade to prevent a hypertensive crisis in pheochromocytoma.

28 Musky the Antagonist
Antimuscarinics

STAT:
M3 Selective



POWER: Secretion Drought
LORE: Tertiary amines (oxybutynin) cross the BBB causing cognitive impairment; quaternary (trospium) do not.

KayfaBOGGLE!

29 Cate the Catecholamine

Adrenaline/Dopamine

STAT

1-3 vs 3-10 vs >10
($\mu\text{g/kg/min}$)



POWER: Receptor Swap

LORE: Noradrenaline restores SVR in septic shock; Dobutamine provides inotropy in cardiogenic shock.

31 Cholinergic vs. Myasthenic

Crisis Matrix

STAT:

Miosis

(The distinguishing sign)



POWER: The Edrophonium Gamble

LORE: Pinpoint pupils distinguish a deadly cholinergic overdose from underlying myasthenic weakness.

30 Anti the Anticholinesterase

AChE Inhibitors

STAT

SLUDGE



POWER: Enzyme Aging

LORE: Atropine rescues the muscarinic effects (secretions) of organophosphates, but does NOT reverse the nicotinic paralysis.

32 Local Anesthetic Lucy

Lidocaine/pH

STAT:

pKa 7.9



POWER: pH Dependence

LORE: Infected, acidic tissue traps local anesthetics outside the nerve, preventing the block.

KayfaBOGGLE!

BLÖDSINN!



33 Tole the Tolerance

Adaptation

STAT
24-48 Hours
(Nitrate tolerance)



POWER: Tachyphylaxis.
LORE: Continuous GTN infusion causes rapid endothelial tolerance; a nitrate-free interval is mandatory.

35 Suzy Synergy

Drug Combinations

STAT:
 $1 + 1 = 3$



POWER: Sequential Blockade
LORE: Co-trimoxazole sequentially blocks two steps of folate synthesis, creating a synergistic bactericidal effect.

34 Anton Antagonist

Competitive/Non-comp

STAT
EC50
Shift Right



POWER: Insurmountable Block
LORE: Fentanyl's massive affinity can overcome standard doses of competitive naloxone, requiring repeat boluses.

36 Thera Therapeutic Index

Safety Margin

STAT:
TD50 / ED50



POWER: The Narrow Window
LORE: NSAIDs + Lithium = reduced renal clearance = coarse tremor and seizures.

KayfaBIZARRO!

37 Indo Induction

General Anesthesia

Analgesia

STAT

MAC %

Hypnosis

Paralysis



POWER: Lipid Solubility

LORE: Ketamine preserves hemodynamics and bronchodilates, but is contraindicated in raised ICP.

38 Prot the Prodrug

Metabolic Activation

STAT

30% PMs
(Clopidogrel resistance)



POWER: Hepatic Activation

LORE: CYP2C19 poor metabolizers suffer stent thrombosis when prescribed clopidogrel.

39 Enter the Enterohepatic

Recycling

STAT:

The Gut Loop



POWER: Half-Life Extension

LORE: Colestyramine binds leflunomide in the gut, permanently breaking the recycling loop for rapid washout.

40 Preg the Pregnancy

Teratogens

STAT:

Weeks 6-12
(Warfarin)



POWER: Teratogenic Strike

LORE: Valproate causes neural tube defects; folic acid supplementation does not reliably prevent them.

Stay Fluffy!

41 NSAID Neil COX Inhibition

STAT
COX-1 / COX-2



POWER: Afferent Constriction
LORE: COX-2 selective inhibitors spare the gastric mucosa but increase cardiovascular pro-thrombotic risk.

42 Proton Petra PPIs

STAT
18 Hours
(Pump synthesis)



POWER: Irreversible Sulfenamide
LORE: PPIs are irreversible but only target active pumps; they must be taken 30-60 minutes before meals.

Med-a-Bizarro!

43 Warry Warfarin VKOR inhibition

STAT:
INR Target
2.0-3.0



POWER: VKOR Blockade
LORE: Factor VII depletes in 6 hours, but full anticoagulation requires Factor II depletion (5 days), necessitating an LMWH bridge.

44 Diuretic Danny Nephron Sites

STAT:
25% Na^+
(Loop Diuretics)



POWER: Ion Wasting
LORE: Loop diuretics cause hypocalcemia; Thiazide diuretics cause hypercalcemia.

HUMBUG!

GW

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45 Statin Steve

HMG-CoA Reductase

STAT

>50% LDL Drop



POWER: Pleiotropic Shield

LORE: Blocking hepatic cholesterol synthesis forces the liver to massively upregulate plasma LDL receptors.

46 Opio the Opioid

MOR Agonists

STAT

10mg IV =
30mg PO
(Morphine)



POWER: Respiratory Ceiling

LORE: Buprenorphine is a partial agonist; it has a respiratory ceiling but will precipitate violent withdrawal if given to heroin users.

47 Antibio the Antibiotic

5 Targets

STAT:

The 5 Vulns



POWER: Target Exploitation

LORE: Oral Vancomycin is not absorbed systemically; it stays entirely in the gut to eradicate *C. difficile*.

48 Time Kill Terry

PK/PD Targets

STAT:

T > MIC vs
C_{max}/MIC



POWER: Post-Antibiotic Effect

LORE: Aminoglycosides are dosed once daily to maximize the C_{max} killing peak while allowing kidneys time to recover.

49 Resi the Resistance

Beta-lactamases/Efflux

STAT

ESBL / KPC



POWER: Horizontal Transfer
LORE: ESBL-producing organisms destroy all cephalosporins; carbapenems are mandatory.

50 Anti-Viral Vera

Viral Lifecycle

STAT

TK
Activation



POWER: Chain Termination
LORE: Aciclovir is a prodrug that is selectively toxic because it can ONLY be activated by viral thymidine kinase.

Med-a-Bizarro!

51 Azole Aziz

Fungal Membrane

STAT:

CYP51
Inhibition



POWER: Sterol Starvation
LORE: Azoles are potent CYP3A4 inhibitors; echinocandins are preferred in ICU patients to avoid massive drug interactions.

52 Chemo Clara

Cell Cycle

STAT:

S-phase
vs. M-phase



POWER: Cycle Arrest
LORE: Cyclophosphamide causes hemorrhagic cystitis via its acrolein metabolite; always protect the bladder with Mesna.

53 Poly Polypharmacy

Elderly/Deprescribing

STAT

ACB Scale



POWER: Cascade Prescribing

LORE: The combination of anticholinergics (confusion) and antihypertensives (orthostasis) is the leading driver of elderly hip fractures.

54 Herb the Herbal

St John's Wort/Grapefruit

STAT

24-72 Hours
(Inhibition)



POWER: Unpredictable Induction

LORE: Grapefruit juice irreversibly inhibits gut CYP3A4; St. John's Wort aggressively induces it.

55 The Attending

Master Review

STAT:

The
Wimmelbild



POWER: Integrated Reasoning

LORE: Map the interaction. Predict the consequence.
State the management. Show your reasoning.

56 The Joker

Wildcard / DIY

STAT:

DIY



POWER: Draw Your Own

LORE: The final mechanism is yours. Leave your mark.